

Flexible Cable Simulation in NVIDIA Isaac Sim

Comparing three modelling approaches for RL-based cable manipulation

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- **Why cables?** wire-harness assembly, cable routing, and deformable-object manipulation are common, unsolved robotics tasks.
- **The challenge:** Isaac Sim has *no native 1-D deformable (rod/cable) body* – only rigid bodies and volumetric FEM soft bodies.
- **The goal:** find the cable model that is realistic *and* fast *and* robot-coupled enough for **RL-based cable manipulation** with domain randomization.

We evaluate three approaches, each in two scenarios (free cable, and cable held by a Franka arm).

Three approaches

1. Capsule chain

rigid capsules + D6 joints; bending from Euler–Bernoulli beam theory (PhysX).

2. Cosserat rod (Warp)

1-D position-based elastic rod in NVIDIA Warp, after the *OGC SIGGRAPH-2025* paper.

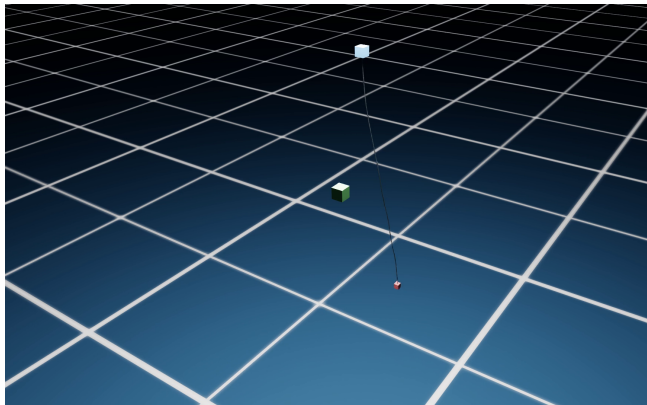
3. FEM deformable

Isaac Sim's built-in volumetric soft body (tetrahedral FEM).

Two scenarios each $\Rightarrow$ **6 simulations**: (A) free hanging/swinging cable manipulator. (B) cable connected to a robot

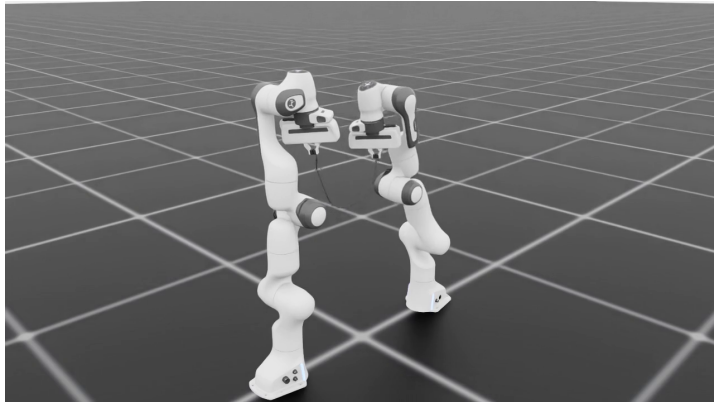
Approach 1 — Capsule chain (D6 joints)

- Rigid capsules linked by **D6 joints**; translations locked (inextensible), bending springs $k = EI/L$ from beam theory.
- PhysX **TGS** solver, up to 200 links, 240 Hz.
- Reproduces the **Govoni et al. 2025** stability sweep.
- Modes: hanging-kick and both-ends-fixed.



Video: *approach1_cable_only/video.mp4* (shared folder)

Approach 1 — With robot

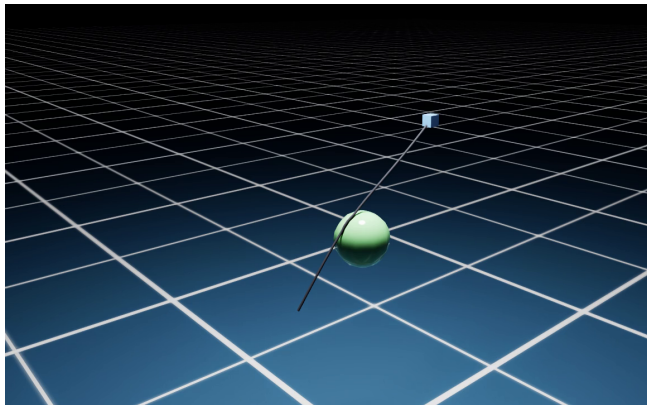


Video: approach1_robot/video.mp4 (shared folder)

Inextensible (rigid joints): a hard pull **drags the second arm** ~ 17 cm while the cable barely stretches (~ 5 cm)
 $\Rightarrow$ **true force transmission.**

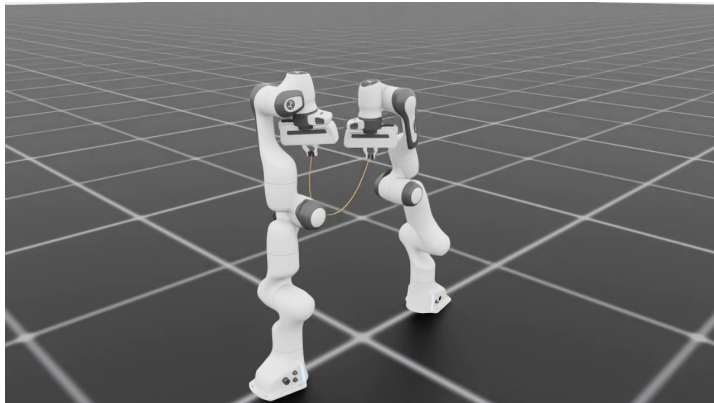
Approach 2 — Cosserat rod in NVIDIA Warp

- 1-D **Cosserat** / **XPBD** elastic rod: inextensible stretch + bending stiffness + gravity, solved on the GPU in **Warp**.
- Idea from the **OGC** paper (Chen et al., SIGGRAPH 2025): penetration-free contact for cloth & rods.
- Truly **thin** (no FEM fattening); **exact** node-vs-primitive collision.



Video: *approach2_cable_only/video.mp4* (shared folder)

Approach 2 — With robot

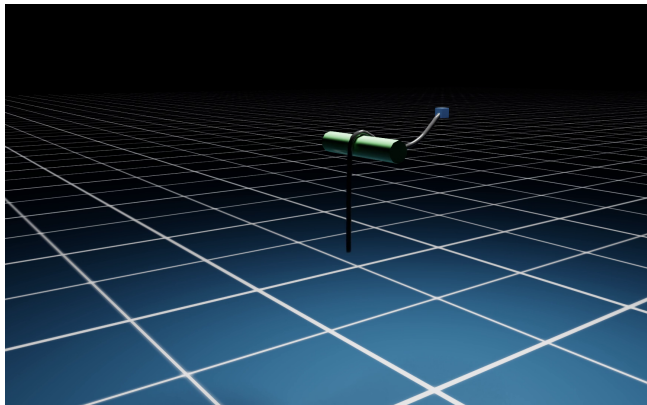


Video: `approach2_robot/video.mp4` (shared folder)

End nodes **pinned** to the grippers: the thin rod follows the arms and sags cleanly between them (best-looking cable). **One-way** – it follows but exerts no force back, so it doesn't drag the second arm.

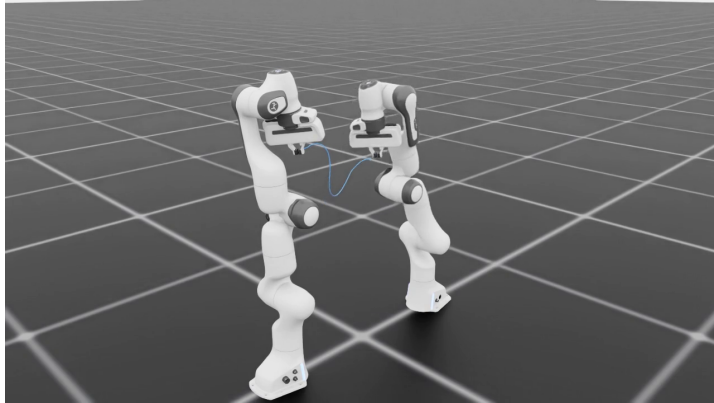
Approach 3 — FEM deformable (Isaac Sim)

- Isaac Sim's built-in **volumetric** (tetrahedral) FEM soft body.
- TPU material: $E = 40 \text{ MPa}$, $\nu = 0.48$, $\rho = 1150 \text{ kg/m}^3$.
- Rod **fattened** to 12 mm and E rescaled so it bends/weights like the real 1.5 mm cable (voxeliser needs ≥ 3 hexes across).
- Rendered thin via a swept tube on its centreline.



Video: *approach3_cable_only/video.mp4* (shared folder)

Approach 3 — With robot



Video: approach3_robot/video.mp4 (shared folder)

Region-weld attachment. Being a soft body it **stretches a lot** under a hard pull ($\sim 20\text{--}35\%$) and drags the follower little $\Rightarrow$ poor force transmission.

Comparison

Criteria	D6 capsule	Cosserat (Warp)	FEM
Realism	Good (beam bending)	Excellent (1-D rod)	Fair (jelly/stretch)
Stability	Excellent	Excellent	Fair
Compute cost	Low (real-time)	Low ($\sim 1.4 \times$ RT)	High ($< 0.2 \times$ RT)
Ease of setup	Moderate	Moderate (custom Warp)	Easy (built-in)
RL readiness	High	Medium	Low
Force transmission	Strong (~ 17 cm)	None (one-way)	Weak (stretches)
Domain randomization	Easy	Easy	Harder

RT = real time. “Two-way” = the robot feels the cable’s reaction force.

Pros and cons

1. Capsule chain (D6 joints)

- + Stable over a wide parameter range; physically grounded; real-time; **two-way** robot coupling; easy domain randomization.
- No axial elasticity (links can't stretch); needs many links for a smooth look; free twist DOF needs damping.

2. Cosserat rod in Warp (OGC)

- + Most faithful cable: thin, inextensible, no jelly; exact, penetration-free collision; GPU-fast; cleanest-looking cable.
- **One-way** robot coupling (follows the grippers but exerts no force back); custom Warp code (not a native Isaac body); no volumetric contact.

3. FEM deformable

- + Most realistic volumetric deformation; handles complex geometry; built into Isaac Sim; two-way.
- Expensive; must be fattened (can't be thin); stretches; rough simplified collision; rigid clamp at the attachment; slow for RL.

For RL-based cable manipulation: the **capsule chain (D6 joints)**

- Only model that is simultaneously **stable, real-time, and two-way robot-coupled** – the robot actually feels the cable, which RL needs.
- Physically grounded (beam theory) and **easy to domain-randomize** (per-joint stiffness/damping/mass).
- Use the **Warp Cosserat rod** as a high-fidelity reference / for visualization and contact-rich studies (penetration-free, exact collision).
- Use **FEM** only when a genuine *volumetric* soft body (squashing/draping) is required – not for fast RL.

- **RL training** of a cable-manipulation policy on the capsule model.
- **Domain randomization**: cable stiffness, damping, mass, length, friction; robot dynamics.
- **Two-way coupling** study: how the cable's reaction shapes the policy.
- **Physical validation** against a real TPU cable.
- Before arriving at Waterloo: finalize the env + reward and a first sim-to-sim transfer between the three models.

Thank you

Questions?

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